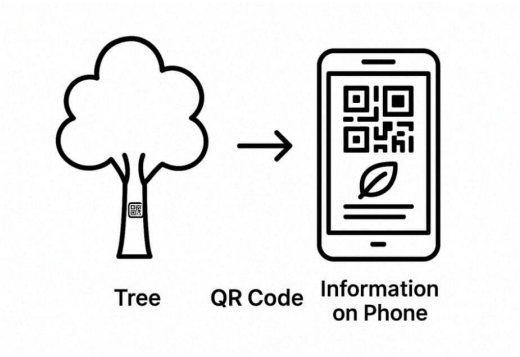
Design Thinking for Sustainability
(Batch 2025-26)

Ideate & Concepts

MIT Vishwaprayag University

Concept Sketches (Hand Drawn/Digital)



Brainstormed Ideas

QR code stickers on plants Which we can Scan to get more information about trees

Why we chose the final concept (Short reasoning)

We selected this concept because it is simple, innovative, and practical. QR code stickers make it easy for anyone to instantly access detailed information about each tree using just a smartphone. This solution is cost-effective, eco-friendly, and interactive, encouraging students and visitors to learn more about biodiversity. It also avoids the need for large physical boards, keeping the forest area natural while still promoting digital learning and awareness.

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Prototype & Testing

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Photographs of User Interacting with Prototype

Testing Process (Who,Where,How)

- **Who: Students, faculty members, and campus visitors tested the QR codes.**
- **Where: QR stickers were placed on selected plants in the Miyawaki forest / campus garden.**
- **How:**
  - **Users scanned QR codes with smartphones.**
  - **Checked if the links opened correctly and provided accurate tree information.**
  - **Collected feedback on ease of use, clarity of information, and user experience.**

User Feedback

- "The QR codes are very easy to scan and give clear information about each tree."
- "I liked how quickly I could learn about the plants without needing a guide."
- "This idea makes the forest more educational and interactive."
- "It's very useful for students as we can get detailed knowledge instantly."

Suggested Improvements

Weatherproof Stickers – Use laminated or metal QR code plates so that rain, dust, and sunlight don't damage them.
Interactive Features – Include quizzes, fun facts, or AR (Augmented Reality) features to engage students more.

Next steps (if implemented at campus scale)

Start with a small section of the forest to test durability of QR codes and collect user feedback..
Gather user feedback to identify improvements.

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Project Title:-Smart Learning Through Miyawaki Forest

Team Member Names:-Sumit, Om, Snehal, Pratik

Problem Overview: Students get confused to identify which plant is it.

Stakeholders (Primary/Secondary): Primary Student & Secondry Worker

Proposed Solution: We have generated QR code with audio text and images

Impact on Sustainability (Linked to SDG):

Know identify the plants name and also get more knowledge about that plant by scanning QR code

Impact (Environmental, Social, Economical)

Educational Value: Students and visitors can scan QR codes to learn about species, making it an outdoor learning lab.Forests with QR codes can attract visitors, researchers.

Prototype Photo

Made A QR Codes To Get Knowledge About Trees

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Discover & Define

User Persona

Name:-Akash
Age:-20
Role : Engineering Student
Needs:-Clear, Accessible Plant identification System
Frustration:-Difficulty identifying tree names without proper nameplates.
Motivation:-Curiosity to learn plant names and their uses instantly.

Empathy Map Findings

Says	Thinks
There are no nameplate on the plants so students are confused that which plant it is	I wish there were labels on plants
Does	**Feel**
Display the nameplates on the plants	**Dissappointed That Learning is incomplete**

Key Insights: The Miyawki Forest Method Enables Rapid Growth In Small Areas Improving Biodiversity Air Quality And Climate Resilience Within Just A Few Years

HMW Statement: How might we Desigh low-cost and sustainable nameplates for plant identification in order to help students learn about bioediversity effectively

Design Objectives: To create a Qr Code for plant identification so student can easily identify plants